



DO-IT-YOURSELF: CIRCUIT IDEA

component layout is in Fig. 4.

After assembling the circuit on the PCB, it should be enclosed in a suitable box, ensuring that the 7-segment displays are positioned on the front panel. The sensor should be extended using a twin-wire connection and installed in a location suitable for the intended application. LED1 and power switch S1 should be fixed on the enclosure's front panel.

After assembly, testing involves powering the system and verifying that the displays count sequentially from 00 to 99. If the displays do not show the expected numbers, possible issues may include wiring faults, incorrect resistor values, or poor connections between the ICs and displays.

EFY note. A debouncing circuit (e.g., an RC circuit) should be added if using manual triggers for the clock.

The total current consumption should not exceed the power supply's capacity.

Each stage (timer, counters, and displays) should be verified before integration.

Modification

The clock switch (S2) can be replaced with a sensor to trigger the LM555 timer (IC1), which will then generate pulses to increment the two-digit counter using CD4033 (IC2 and IC3).

A sensor that produces a momentary high-to-low clock signal can be used instead of a mechanical switch. Suitable options include:

- IR proximity sensor (for object detection)
- LDR with comparator LM358 (for light-based counting)
- PIR motion sensor (for motion-based triggering)

After designing, the sensor should be mounted at the counting point (e.g., where an object passes, a light beam is interrupted, or a hand moves). It should be wired in place of the switch so that each activation sends a pulse to the 555 timer. An oscilloscope or an LED indicator can verify that the sensor produces a clean pulse. The CD4033 should correctly increment the display when the sensor is triggered. **EFY**

Bonus. You can watch the video of the tutorial of this DIY project at:

<https://www.electronicsforu.com/videos-slideshows/live-diy-make-your-own-two-digit-counter>

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